

Distribution Fault-Locating Algorithms Using Current Only

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Abstract—Traditional impedance-based fault-locating methods implemented in modern overcurrent protection relays require voltage and current measurements to provide reasonable fault-location estimates. Although they capture voltage and current, depending on field condition or due to equipment failure, relays may record current measurements only. Voltage measurements are thus missing or unavailable. The objective of this paper is to develop practical impedance-based fault-locating algorithms with current data (magnitude or phasors) as the only input and demonstrate the efficacy of the algorithms with simulated and actual field data. These algorithms use the circuit model of the distribution feeder and Kirchhoff's circuit laws in estimating the fault voltage at the relay location and then use impedance-based methods for fault location. Based on the analysis conducted on actual fault data, error in estimation is generally less than 0.5 mi from the actual location of the fault.

Index Terms—Fault current, fault location, power distribution faults, power system faults, power system reliability.

I. INTRODUCTION

IMPEDANCE-BASED fault-locating methods, such as those of Takagi and loop reactance, have been adopted for transmission and distribution circuit applications. They are straightforward to implement and yield reasonable location estimates. The algorithms require the availability of voltage and current data captured when the fault is on the circuit [1]–[3]. Modern digital relays can record voltage and current waveforms. However, depending on relay settings and other operational considerations or constraints, a relay may record current phasor (magnitude and phase-angle quantities) or current magnitude (no phase angle) only. Voltage waveforms are thus missing, unavailable, or simply not recorded.

The objective of this paper is to use the existing impedance-based fault-locating algorithms implemented in commercial relay software and extend the application of these algorithms with current data as the only input. The study presented in this paper is part of an Electric Power Research Institute (EPRI) fault-location study and demonstrates the efficacy of the proposed algorithms with simulated and actual field data. The goal is to locate single-line-to-ground faults since they are

the most common among all other types of faults. The proposed algorithms are novel and have not been demonstrated in the literature. An alternative to the current-only impedance-based approach is the current profile location method described in [3]–[5]. To obtain a location estimate, actual fault currents are matched with known short-circuit currents available from the circuit model of the distribution feeder. A similar problem is explored in [6] and [7], wherein an impedance-based matrix technique is used to locate a fault using only current magnitude and/or phasor measurement from one or more buses in the network. The approach assumes that the system is completely balanced and has been tested on a trivial four-bus test system. The algorithms proposed in this paper are based on a single-point measurement, and the system does not have to be balanced. They have also been validated using real-world data with known fault locations.

Depending on data recorded by a relay (current phasor or magnitude), the algorithms have been developed in two parts: fault location using current phasors and fault location using current magnitude only. Both approaches require the circuit model of the distribution feeder. In addition, the algorithm using current phasors requires prefault current data. In both approaches, the fault voltage at the monitoring location is first estimated using Kirchhoff's circuit laws and subsequently an impedance-based method is applied to estimate distance to the fault. These algorithms are computationally efficient and, hence, voltage estimation as well as fault-location estimation can be implemented within the relay. A description of the approaches and algorithm derivation are presented in Sections II and III, respectively.

In Section IV, the algorithms are initially evaluated using the known circuit topology of the IEEE 34-node test feeder. The feeder was modified to represent difficult and challenging scenarios of fault locating. Despite the worst-case scenarios, the algorithms using current phasors and magnitude gave accurate estimates. The highest error in estimation was observed for faults at the farthest buses where the load current was nearly 50% of the total fault current seen by the relay. The algorithms were then evaluated using seven actual fault case events collected from distribution circuits, as described in Section V. The maximum error observed is 4.5% (0.34 mi) and 8% (0.61 mi) in the calculated fault location using current phasors and current magnitude, respectively.

II. APPROACH AND APPLICATION SCENARIO

When a single-line-to-ground fault occurs in the distribution feeder, the relay at the monitoring location may record only the fault current phasor or the fault current magnitude. Hence, the

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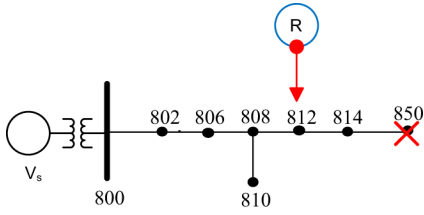


Fig. 1. One-line diagram of the example test case.

approach to fault location using only current has been developed in two parts.

A. Fault Location Using Only Current Phasors

The relay records rms magnitude and phase angle of the pre-fault as well as fault current. Using the circuit model of distribution feeder, fault voltage at the monitoring location is derived using simple Kirchhoff's circuit laws and then impedance-based methods, such as the Takagi method, are used to estimate the distance to the fault. Prefault current phase angle is required as a reference to accurately determine the phase angle of fault voltage.

B. Fault Location Using Only Current Magnitude

In this approach, only magnitude of fault current is available. No prefault current is required. The circuit model of the distribution feeder and Kirchhoff's circuit laws are used to estimate the magnitude of fault voltage at the monitoring location. The loop reactance method is then used to estimate fault location since it has a simple formula which can be modified to work with only the magnitude of fault current and voltage.

C. Application Scenario

A scenario for this approach may be illustrated as follows. A single-line-to-ground fault occurs in the distribution system. The relay upstream from the fault location records only the current waveform (from which magnitude and phase angle can be determined) or only the current magnitude. The utility has the circuit model of the distribution feeder in one of the distribution system software packages. Based on the information obtained from the relay, the appropriate algorithm is applied to determine the location of the fault.

III. DERIVATION OF ALGORITHMS FOR FAULT LOCATION USING CURRENT ONLY

For the derivation of the algorithms, a small section of the IEEE 34-node test feeder shown in Fig. 1 is considered as an example case [8]. A single-line-to-ground fault occurs at bus 850. The relay is located at bus 812.

A. Fault Location Using Current Phasors

This section gives a step-by-step derivation of the algorithm, illustrating how the utility can use current phasor as the only data input to determine the location of a fault.

1) *Thevenin Equivalent at the Relay Terminals:* The circuit upstream from the relay is represented by a Thevenin equivalent

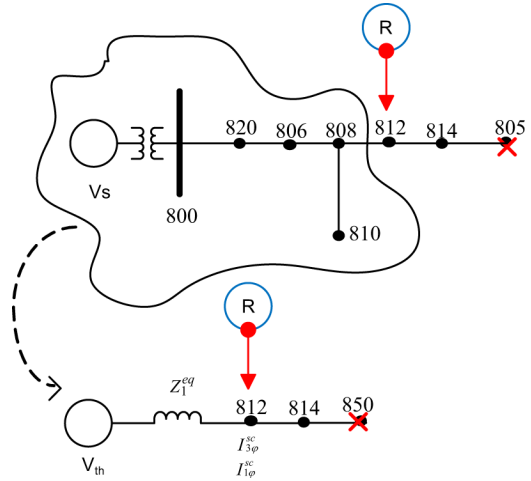


Fig. 2. Circuit upstream from the relay monitoring location represented by a Thevenin equivalent circuit.

circuit which consists of an ideal voltage source behind equivalent impedance shown in Fig. 2. The positive- and zero-sequence equivalent impedance Z_1^{eq} and Z_0^{eq} can be calculated as [9]

$$Z_1^{\text{eq}} = \frac{kV_{LL}}{\sqrt{3} \times I_{3\phi}^{\text{sc}}} \Omega \quad (1)$$

$$Z_0^{\text{eq}} = \frac{\sqrt{3} \times kV_{LL}}{I_{1\phi}^{\text{sc}}} - 2Z_1^{\text{eq}} \Omega \quad (2)$$

where kV_{LL} is the rated line-to-line voltage (in kilovolts) and $I_{3\phi}^{\text{sc}}$ and $I_{1\phi}^{\text{sc}}$ are the fault currents (in kiloamperes) of a three-phase and single-phase short circuit at the relay location. Since the utility has the circuit model of the distribution feeder, the short-circuit currents can be easily obtained. The Thevenin equivalent voltage (V_{th}) is the open-circuit voltage at the relay terminals. In any practical power system, it is close to the source voltage [9]. The phase angle of V_{th} ($\theta_{v_{\text{th}}}$) is calculated from the power factor (pf) of the circuit. The power factor can be determined by carrying out a load-flow analysis on the circuit model of the distribution feeder. Using this power factor value, $\theta_{v_{\text{th}}}$ can be calculated as

$$\cos(\theta_{v_{\text{th}}} - \theta_{i, \text{prefault}}) = pf \quad (3)$$

where $\theta_{i, \text{prefault}}$ is the phase angle of the prefault current. Therefore

$$\theta_{v_{\text{th}}} = \cos^{-1}(pf) + \theta_{i, \text{prefault}} \quad (4)$$

In the subsequent subsections, $\theta_{v_{\text{th}}}$ will be used in Kirchhoff's circuit law equations to estimate the fault voltage phasor at the relay location. Hence, the accuracy of $\theta_{v_{\text{th}}}$ is important to obtain an accurate fault-location estimate.

2) *Sequence Networks During the Fault:* In terms of the sequence networks, the example circuit in Fig. 2 during the fault can be represented as shown in Fig. 3[10]. The sequence network has been drawn for a positive-sequence network. Currents I_{f1} , I_{f2} , and I_{f0} are the sequence components of fault current

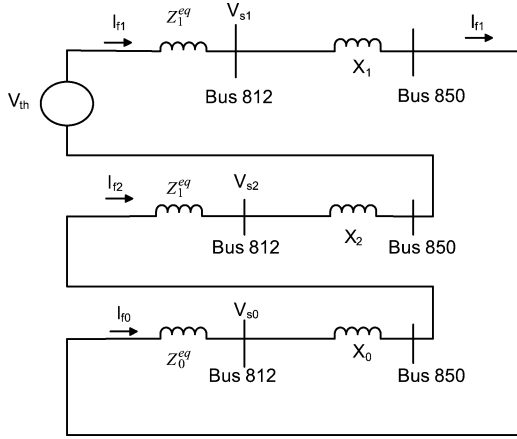


Fig. 3. Sequence network of the example test case. The monitoring location is at bus 812.

recorded by the relay; X_1 , X_2 , and X_0 are the unknown sequence reactances from the monitoring location to fault and V_{s1} , V_{s2} , and V_{s0} are the sequence components of the fault voltage at the monitoring location.

3) *Determining the Fault Voltage at the Monitoring Location:* From the sequence network, using simple Kirchhoff's circuit laws, the sequence components of the fault voltage at the monitoring location can be written as

$$V_{s1} = V_{th} - I_{f1} \times Z_1^{eq} \quad (5)$$

$$V_{s2} = 0 - I_{f2} \times Z_2^{eq} \quad (6)$$

$$V_{s0} = 0 - I_{f0} \times Z_0^{eq}. \quad (7)$$

The fault voltage at the monitoring location ($V_{monitor}$) is then given by

$$V_{monitor} = V_{s1} + V_{s2} + V_{s0}. \quad (8)$$

4) *Application of Impedance-Based Algorithms:* Once the missing fault voltage has been estimated using (8), impedance-based methods can be used to calculate the reactance or the distance to the fault. In this paper, the Takagi method has been used. The Takagi method minimizes the effect of fault resistance [2], [3] and estimates the distance to fault (d) as

$$d = \frac{\text{Im}(V_{monitor} I_{sup}^*)}{\text{Im}(z_{line,1} I_s I_{sup}^*)} \quad (9)$$

$$I_{sup} = I - I_{preft} \quad (10)$$

where $V_{monitor}$ is the estimated fault phase voltage at the relay (in kilovolts), I is the faulted phase current at the relay (in kiloamperes), I_{preft} is the pre-fault load current in the faulted phase (in kiloamperes) and $z_{line,1}$ is the positive-sequence distribution line impedance (Ω/mile). Note that the Takagi method assumes that the distribution feeder has a homogeneous line configuration. In practice, distribution feeders have different line configurations. In such a case, $z_{line,1}$ of the most commonly occurring line configurations should be used [11].

B. Fault Location Using Current Magnitude

In the previous subsection, the approach is to use the fault current phasors recorded by the relay and estimate the missing fault

voltage phasor using Kirchhoff's circuit laws. Impedance-based methods can then be used to obtain a location estimate. When the relay records only the fault current magnitude, the aforementioned approach can be extended with additional modifications. This section illustrates the derivation of the algorithm wherein the utility can use fault current magnitude as the only data input for fault location. The modifications involved and the rationale behind implementing those modifications have been discussed in detail.

1) *Determining the Sequence Components of Fault Current:* Suppose that a fault has occurred in phase A. The current phasors flowing through in each phase during the fault are I_a , I_b , and I_c . When calculating the symmetrical components, phasor values of I_a , I_b , and I_c are required [10]. Since only the magnitudes of I_a , I_b , and I_c are available, constraints are imposed on calculating the sequence currents. A simplifying assumption is made to ignore the load currents in the unfaulted phases, since $|I_b|$ and $|I_c| \ll |I_a|$. Therefore, symmetrical components of fault currents with respect to the A-phase are

$$\begin{bmatrix} I_{f0} \\ I_{f1} \\ I_{f2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

$$I_{f0} = I_{f1} = I_{f2} = \frac{I_a}{3}. \quad (12)$$

In polar form, the following equation is obtained:

$$I_{f0} = I_{f1} = I_{f2} = \frac{|I_a|}{3} \angle \theta. \quad (13)$$

Here, θ is the angle of the faulted phase current. Taking the magnitude only

$$|I_{f0}| = |I_{f1}| = |I_{f2}| = \frac{|I_a|}{3}. \quad (14)$$

2) *Determining the Thevenin Equivalent:* Equivalent Thevenin impedance Z_1^{eq} and Z_0^{eq} can be calculated from (1) and (2). The Thevenin equivalent voltage (V_{th}) is equal to the source voltage. Since this algorithm works with only magnitudes, there is no need to calculate θ_{vth} .

3) *Estimating Fault Voltage at the Monitoring Location:* After estimating the sequence fault currents, Kirchhoff's circuit laws can be used to solve for the missing fault voltage. However as given by (5)–(8), these equations are designed to work with only phasor quantities and have to be modified to work with only magnitudes of I_{f1} , I_{f2} , and I_{f0} . For this purpose, (5) is rewritten in polar form as

$$|V_{s1}| \angle \theta_{v_{s1}} = |V_{th}| \angle \theta_{vth} - (|I_{f1}| \angle \theta_{f1} \times |Z_1^{eq}| \angle \theta_{z_1}) \quad (15)$$

where $\theta_{v_{s1}}$ is the phase angle of the positive-sequence fault voltage at the relay, θ_{f1} is the phase angle of the positive-sequence fault current recorded by the relay, and θ_{z_1} is the phase angle of the equivalent Thevenin impedance Z_1^{eq} . Rearranging the terms in (15)

$$|V_{s1}| \angle \theta_{v_{s1}} = |V_{th}| \angle \theta_{vth} - (|I_{f1}| \times |Z_1^{eq}|) \angle (\theta_{f1} + \theta_{z_1}). \quad (16)$$

Taking the magnitude of (16)

$$|V_{s1}| = |V_{th}| \angle \theta_{vth} - (|I_{f1}| \times |Z_1^{eq}|) \angle (\theta_{f1} + \theta_{z_1}). \quad (17)$$

From the reverse triangular inequality theorem [12]

$$|u - v| \geq ||u| - |v||. \quad (18)$$

Therefore, (17) can be written as

$$|V_{s1}| \geq |V_{th}| - (|I_{f1}| \times |Z_1^{eq}|). \quad (19)$$

Now, the equality condition $|V_{s1}| = |V_{th}| - (|I_{f1}| \times |Z_1^{eq}|)$ will hold true only if

$$\angle \theta_{vth} = \angle (\theta_{z1} + \theta_{f1}). \quad (20)$$

Assuming condition (20) to hold true, the sequence fault voltages at the relay location in terms of magnitude can be written as

$$|V_{s1}| = |V_{th}| - (|I_{f1}| \times |Z_1^{eq}|) \quad (21)$$

$$|V_{s2}| = -|I_{f2}| \times |Z_2^{eq}| \quad (22)$$

$$|V_{s0}| = -|I_{f0}| \times |Z_0^{eq}|. \quad (23)$$

Summing up the sequence components, the magnitude of fault voltage at relay location ($V_{monitor}$) is given by

$$V_{monitor} = |V_{s1}| + |V_{s2}| + |V_{s0}|. \quad (24)$$

4) *Determining Error if θ_{vth} is not Equal to $(\theta_{z1} + \theta_{f1})$:* Condition (20) requires θ_{vth} to be equal to $(\theta_{z1} + \theta_{f1})$. For an ideal, balanced distribution feeder, this condition is generally fulfilled. However, as will be demonstrated in Sections IV and V, distribution feeders have single-phase or two-phase laterals and single-phase loads, making the feeder potentially unbalanced. As a result, θ_{vth} is not exactly equal but very close to $(\theta_{z1} + \theta_{f1})$. To determine how this error will affect the fault voltage estimation, writing the right-hand side (RHS) of (16) in rectangular form

$$|V_{s1}| \angle \theta_{v_{s1}} = |V_{th}| (\cos \theta_{vth} + j \sin \theta_{vth}) - (|I_{f1}| \times |Z_1^{eq}|) (\cos(\theta_{f1} + \theta_{z1}) + j \sin(\theta_{f1} + \theta_{z1})). \quad (25)$$

Rearranging the real and imaginary terms in the right-hand side of (25)

$$\begin{aligned} |V_{s1}| \angle \theta_{v_{s1}} &= |V_{th}| \cos \theta_{vth} \\ &\quad - (|I_{f1}| \times |Z_1^{eq}|) \cos(\theta_{f1} + \theta_{z1}) \\ &\quad + j(|V_{th}| \sin \theta_{vth} - (|I_{f1}| \times |Z_1^{eq}|) \\ &\quad \times \sin(\theta_{f1} + \theta_{z1})). \end{aligned} \quad (26)$$

Equating only the magnitudes of the left-hand side and right-hand side in (26)

$$|V_{s1}| = \sqrt{|V_{th}|^2 + |I_{f1}|^2 |Z_1^{eq}|^2 - 2|V_{th}||I_{f1}||Z_1^{eq}| \cos(\theta_{vth} - (\theta_{z1} + \theta_{f1}))}. \quad (27)$$

Now, if the equality assumption holds true (i.e., θ_{vth} is equal to $(\theta_{z1} + \theta_{f1})$), $\cos(\theta_{vth} - (\theta_{z1} + \theta_{f1})) = 1$ and, hence, (21) is obtained. However, if they are not equal, the cosine of the angle difference between θ_{vth} and $(\theta_{z1} + \theta_{f1})$ will take a value between 0 and 1. As a result, the actual value of $|V_{s1}|$ will be

greater than the estimated value of $|V_{s1}|$, calculated using (21). In other words, because of the angle mismatch, $|V_{s1}|$ will be underestimated, and the error will be proportional to the square root of the cosine of the difference between θ_{vth} and $(\theta_{z1} + \theta_{f1})$. As will be analyzed in Section IV, this error is not substantial and a difference of 40° is considered to be within acceptable limits.

5) *Distance Estimation Using the Loop Impedance Method:* Like the other impedance-based methods, the loop impedance method also requires phase angles of the fault voltage and current to accurately compute fault distance. However because of its simplicity, it can be adapted to work with only the magnitudes of those phasor quantities. Consider a single-line-to-ground fault ($R_f = 0 \Omega$) in the feeder. The loop impedance (Z_n) to the fault location, as described in [2] and [3], is

$$Z_n = \frac{V_{monitor}}{3 \times I_{f0}}. \quad (28)$$

Writing (28) in polar form

$$|Z_n| \angle \theta_z = \frac{|V_{monitor}| \angle \theta_v}{3 \times |I_{f0}| \angle \theta_i} = \frac{|V_{monitor}| \angle (\theta_v - \theta_i)}{3 \times |I_{f0}|} \quad (29)$$

where θ_z is the phase angle of the loop impedance, θ_v is the phase angle of the fault voltage, and θ_i is the phase angle of the fault current. Another alternative method to calculate loop impedance Z_n is

$$Z_n = \frac{(2d \times z_{line,1}) + (d \times z_{line,0})}{3} = d(r_s + jx_s) \Omega \quad (30)$$

where d is the distance to fault (in miles), $z_{line,1}$ and $z_{line,0}$ are the positive- and zero-sequence line impedance of the most commonly occurring line configuration (Ω/mi), and r_s and x_s are the loop resistance and reactance (Ω/mile). The magnitude and phase angle of (30) are given by

$$|Z_n| = \sqrt{d^2(r_s^2 + x_s^2)} \Omega \quad (31)$$

$$\theta_z = \tan^{-1} \frac{x_s}{r_s} \circ. \quad (32)$$

Substituting (31) and (32) in (29) and equating the magnitude terms, the distance to fault can be determined as

$$d = \frac{\left(\frac{|V_{monitor}|}{3 \times |I_{f0}|} \right)}{\sqrt{r_s^2 + x_s^2}}. \quad (33)$$

Note that the distance to fault in (33) has been derived for a bolted fault (i.e., no fault arc resistance). This method will work best in identifying most distribution faults since their fault impedances are normally very low [3].

IV. APPLICATION OF FAULT-LOCATING ALGORITHMS USING CURRENT ONLY ON SIMULATED FAULT DATA

To evaluate the efficacy of the algorithms developed in Section III, the circuit topology of the IEEE 34-node test feeder was adopted [8]. This feeder was adopted because of the complex topology of the feeder and a wide variety of components which make it an ideal system for fault-location analysis. The feeder is extensive with a complete dataset about the load, conductor type, and other apparatus, making it a useful benchmark.

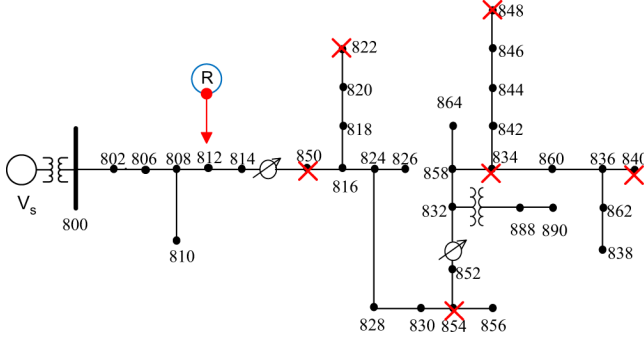


Fig. 4. Fault locations on the test feeder.

A. Modification of the Test Feeder

The test feeder is modeled in PSCAD simulation software using data in [8] with some additional modifications. The ideal 24.9-kV voltage source at the substation is replaced by a 115-kV ideal voltage source stepped down to 24.9 kV by a step-down transformer with a leakage reactance of 0.04 p.u., as shown in Fig. 4. Next, the load current is increased from 40 to 425 A per phase. The fault current is increased to 5 kA (for faults closer to the substation) and 970 A (for faults at a maximum distance from the substation) by decreasing the length of the three-phase branches 812–840 and 812–848. The total length of the modified feeder is 9.53 mi. These modifications are intended to stress test the algorithms.

B. Application of the Fault-Locating Algorithms

Single-line-to-ground faults ($R_f = 0 \Omega$) have been staged at six strategic locations on the test feeder, as shown in Fig. 4. The relay at bus 812 records either a fault current waveform or fault current magnitude. Using the guidelines outlined in Section III, fault locations are estimated as follows.

1) *Fault Location Using Current Phasors*: In this approach, assume that the relay at bus 812 records only the current waveform at 128 samples/cycle. To obtain the rms magnitude and phase angle of prefault and fault current, the classical Fast Fourier transform (FFT) approach is used. The FFT operation isolates only the fundamental frequency and filters out the dc offset. To avoid measurement errors in fault current phasor due to current-transformer (CT) saturation, the dc offset is allowed to decay (in one or two cycles), and the FFT is then performed on the sixth cycle of the fault current. This is due to the fact that CT saturation is associated with fault currents having large dc offsets [3].

Next, to determine the Thevenin equivalent upstream from the relay, three-phase and single-phase faults are simulated in the circuit model. The measured short-circuit currents are

$$I_{3\phi}^{sc} = 4.59 \angle -162.59^\circ \text{ kA}$$

$$I_{1\phi}^{sc} = 3.56 \angle -171.95^\circ \text{ kA}.$$

Using (1) and (2), the equivalent positive- and zero-sequence Thevenin impedance are calculated to be $Z_1^{eq} = 0.1518 + j0.1306$ p.u. and $Z_0^{eq} = 0.2102 + j0.2965$ p.u. on a 40-MVA base. Since it is a positive-sequence circuit (ABC rotation), (5)–(8) are used to estimate the fault voltage at the

TABLE I
COMPARISON OF ACTUAL AND ESTIMATED FAULT VOLTAGE AT THE RELAY:
CURRENT PHASOR APPROACH

Fault Bus	Fault Voltage at the Relay (pu)		Error (%)
	Actual Fault Voltage	Estimated Fault Voltage	
850	0.22	0.25	3.00
854	0.56	0.56	0.00
822	0.68	0.66	2.00
834	0.73	0.70	3.00
840	0.83	0.79	4.00
848	0.84	0.80	4.00

TABLE II
COMPARISON OF THE ACTUAL AND ESTIMATED DISTANCE TO FAULT
USING CURRENT PHASORS

Fault Bus	Distance to Fault			Fault to Load Current Ratio
	Actual (miles)	Estimate (miles)	Error (%)	
850	0.62	0.67	0.52	7.0
854	2.18	2.06	1.26	4.3
822	3.00	2.59	4.30	3.4
834	4.05	3.70	3.67	3.0
840	6.75	5.69	11.12	2.2
848	7.47	6.16	13.75	2.0

monitoring location. Thevenin voltage V_{th} at the monitoring location is equal to the source voltage and is chosen to be 1 p.u. Load-flow analysis on the circuit model indicates that the feeder is operating at a 0.939 power factor. Angle θ_{vth} can then be calculated using (4). The results of voltage estimation at bus 812 are tabulated in Table I. The estimated fault voltages are close to the actual fault voltage with a maximum error of 4%. It should be noted that the error percentage has been calculated with respect to the system nominal voltage (24.9 kV).

After estimating the voltage phasor, the values of $z_{line,0}$ and $z_{line,1}$ are required to estimate distance to fault using the Takagi method. The IEEE 34-node test feeder has five different line configurations, namely, 300, 301, 302, 303, and 304. Since 301 is used for the majority length of the feeder, sequence impedances of this configuration (i.e., $z_{line,1} = 1.6901 + j0.8412 \Omega/\text{mi}$ and $z_{line,0} = 2.3875 + j2.5782 \Omega/\text{mi}$) are used to estimate the distance to fault. The actual and estimated distance to fault is shown in Table II. Fault-location error % is calculated as [1]

$$\text{Error (\%)} = \frac{|\text{Estimated Distance} - \text{Actual Distance}|}{\text{Total Length of the Feeder}} \times 100. \quad (34)$$

As seen from Table II, the error in fault-location estimation increases with the distance from the relay. At the farthest bus (848), the estimation error is 1.31 mi or 13.75%. This is due to the fact that as the distance from the relay increases, the fault current decreases. As a result, the ratio of fault current to load current is around 2 (i.e., load current is nearly 40% to 50% of the total fault current seen by the relay), which affects location

TABLE III
COMPARISON OF ACTUAL AND ESTIMATED FAULT VOLTAGE AT THE RELAY:
CURRENT MAGNITUDE APPROACH

Fault Bus	Fault Voltage at the Relay (pu.)		Error (%)
	Actual Fault Voltage	Estimated Fault Voltage	
850	0.23	0.22	1.00
854	0.56	0.52	4.00
822	0.68	0.62	6.00
834	0.73	0.67	6.00
840	0.83	0.76	7.00
848	0.84	0.77	7.00

estimates [13]. It must be noted that the estimates obtained are conservative since the test feeder has been modeled to represent the worst case wherein the load current has been intentionally kept high. Another important observation is that for some cases, multiple locations are estimated. The IEEE 34-node test feeder, as a radial circuit with many branches and laterals, has many points that exist which fulfill the equivalent distance condition from the relay. For example, for the fault on bus 822, the distance to fault is estimated to be 2.59 mi from the relay. When tracing this estimated distance on the system map, the 2.59 mi may be located on the 816–822 lateral or on the main branch 816–824. An operator can narrow the choices by using a customer outage report, recloser operation status, or field experience to determine which location has the maximum probability of fault.

2) *Fault Location Using Current Magnitude:* In this approach, the relay at bus 812 records only the fault current magnitude. The first step is to calculate the sequence components of the fault current using the assumption made in (11). Next, the Thevenin voltage V_{th} is determined to be 1 p.u. Using (21)–(24), the fault voltage magnitude is computed and shown in Table III. Note that the error has been calculated with respect to the system nominal voltage (24.9 kV). As observed in Table III, for a fault at a particular bus, the fault voltage at the relay is underestimated. Also comparing Tables I and III, the error in estimating the fault voltage is higher in this approach. The additional error is introduced due to the two assumptions given by (11) and (20). The first assumption in (11) ignores the current in the unfaulted phases when computing the magnitude of the sequence fault currents. For example, consider bus 850. The fault current is 7 times the current in the unfaulted phases, as given by the fault-to-load current ratio in Table V. Using (14), the sequence currents are estimated as

$$|I_{f0}| = |I_{f1}| = |I_{f2}| = 901.3 \text{ A.}$$

The actual sequence components of fault current recorded by the monitor are

$$\begin{aligned} |I_{f0,\text{actual}}| &= 738.4 \text{ A} \\ |I_{f1,\text{actual}}| &= 1174.9 \text{ A} \\ |I_{f2,\text{actual}}| &= 795.4 \text{ A.} \end{aligned}$$

TABLE IV
EQUALITY CONSTRAINT CHECK AT EVERY BUS

Fault Bus	$\theta_{v_{th}}$ (degrees)	$\theta_{z1} + \theta_{f1}$ (degrees)
850	-124.59	-122.29
854	-124.59	-116.29
822	-124.59	-119.29
834	-124.59	-116.29
840	-124.59	-115.29
848	-124.59	-115.29

TABLE V
COMPARISON OF ACTUAL AND ESTIMATED DISTANCE TO FAULT USING
CURRENT MAGNITUDE

Fault Bus	Distance to Fault			Fault to Load Current Ratio
	Actual (miles)	Estimate (miles)	Error (%)	
850	0.62	0.47	1.57	7.0
854	2.18	1.82	3.78	4.3
822	3.00	2.71	3.04	3.4
834	4.05	3.37	7.14	3.0
840	6.75	5.18	16.47	2.2
848	7.47	5.53	20.36	2.0

The estimated sequence currents are close to the actual sequence currents, and the error is calculated as

$$\text{Error \%} = \frac{|\text{Estimated Value} - \text{Actual Value}|}{\text{Actual Value}} \times 100. \quad (35)$$

Therefore, the assumption in (11) results in an error of 22.1% when estimating I_{f0} and an error of 23.3% and 13.3% when estimating I_{f1} and I_{f2} , respectively. As seen in Table III, this does not cause a significant error percentage in estimating the fault voltage at bus 850. However, for faults at buses 840 and 848, currents in the unfaulted phases are nearly 40% to 50% of the fault current. As a result, the estimated sequence currents under this assumption have a higher error percentage with respect to the actual sequence current. Because of this high error percentage, the voltage estimated at the relay for the faults at buses 840 and 848 has a greater error percentage of 6 to 7%, as shown in Table III. The lesson learned at the end of this analysis is that the performance of fault location using the current magnitude approach is superior when fault current is greater than twice the current in the unfaulted phases.

The next assumption that contributes to the error in fault voltage estimation is the equality constraint given by (20). As shown in Table IV, at each bus, $\theta_{v_{th}}$ is not exactly equal to $(\theta_{z1} + \theta_{f1})$. The angle mismatch is maximum (9.3°) at bus 848. To calculate the error percentage using (27), consider the simulation values for bus 848

$$|V_{th}| = 1 \text{ p.u.}, |I_{f1}| = 0.266 \text{ kA and } |Z_1^{\text{eq}}| = 3.1 \Omega.$$

For $\theta_{v_{th}} - (\theta_{z1} + \theta_{f1}) = 9.3^\circ$, the actual value of $|V_{s1}| = 13.56 \text{ kV}$. However, because of the equality assumption (i.e., $\theta_{v_{th}} = (\theta_{z1} + \theta_{f1})$), $|V_{s1}|$ is estimated to be 13.55 kV. The error in estimation is negligible, around 0.1%. Next, to preserve

the accuracy level for calculating $|V_{s1}|$, it is necessary to find an acceptable limit for this angle mismatch. For this purpose, the difference between θ_{vth} and $(\theta_{z1} + \theta_{f1})$ is gradually incremented in (27). Simulation values of bus 848 have been used. It was observed that for a difference of 40° , the error in estimating $|V_{s1}|$ is within the 2% tolerance level. Since the IEEE 34-node test feeder is representative of a real-world feeder, it was concluded from this analysis that a 40° mismatch between θ_{vth} and $(\theta_{z1} + \theta_{f1})$ is acceptable for fault voltage calculation.

Using the estimated fault voltage in Table III, the modified loop reactance derived in (33) is used to estimate the distance to fault. Line impedance parameters of configuration 301 are used. As shown in Table V, error in estimation is within 1.94 mi. Recall that the test feeder has been modeled to represent the worst case scenario wherein the load current is very high (400+ A). The loop reactance method does not take into account load current and the estimates obtained are hence conservative [13].

V. APPLICATION OF FAULT-LOCATING ALGORITHMS USING CURRENT ONLY ON FIELD DATA

In this section, the application of fault-locating algorithms using current only is demonstrated on seven actual fault events. Each event has monitoring data and a system circuit model. Fault location on two of the events has been explained in detail followed by a summary of the remaining events.

A. Utility Test Case 1

The circuit model has been made available by a utility as part of the EPRI distribution fault-location study. The rated voltage at the substation is 23.9 kV. A SEL-351S relay located at the substation monitors the system current and voltage at 16 samples/cycle. Since the relay is located at the substation, the Thevenin equivalent positive- and zero-sequence impedance upstream from the relay is equal to the source impedance and can be obtained from the circuit model as $Z_1^{eq} = 2.453 + j44.73$ p.u. and $Z_0^{eq} = 1.16 + j39.12$ p.u. The length of the main three-phase feeder is 7.61 mi and is constructed using three different three-phase line configurations. The sequence impedance of the most commonly occurring conductor type in the circuit (i.e., $z_{line,1} = 0.1680 + j0.6218$ Ω /mi and $z_{line,0} = 0.4372 + j1.9305$ Ω /mi) has been utilized for computing the distance to fault. A C-phase line-to-ground fault has occurred in a lightning arrester located 5.33 mi from the substation due to squirrel contact, as shown in Fig. 5. To estimate the location of the fault, the algorithms developed in Section III were applied.

1) *Fault Location Using Current Phasors:* For the analysis with fault current phasor, pretend that the relay records only the current waveform as shown in Fig. 6. The FFT operation has been performed on this waveform to extract the fundamental frequency prefault and fault current of phase C as

$$I_{prefault,phaseC} = 200 \angle -34^\circ \text{ A}$$

$$I_{fault,phaseC} = 1870 \angle -93^\circ \text{ A.}$$

To avoid errors due to dc offset and CT saturation when calculating $I_{fault,phaseC}$, the fault current is allowed to reach steady

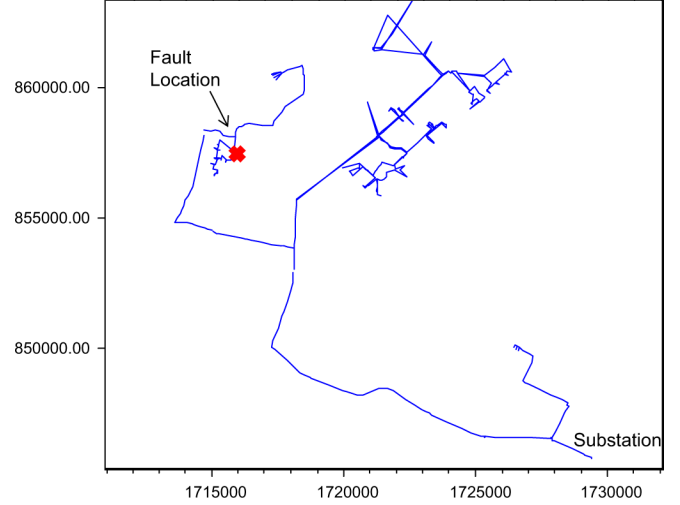


Fig. 5. Location of the fault in the feeder.

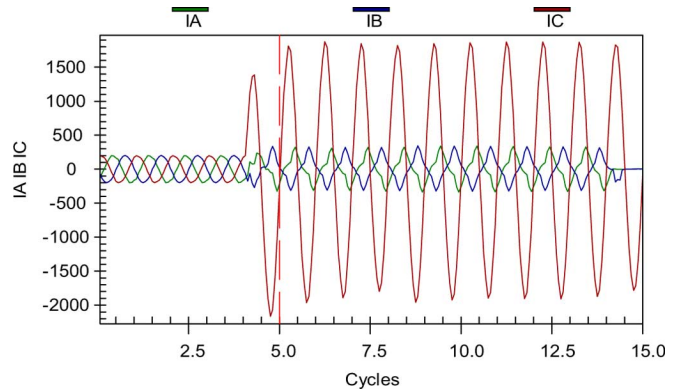


Fig. 6. Relay at the substation recording the prefault and fault current waveform.

state and the FFT operation has been performed on the sixth cycle after the initiation of the fault.

From the circuit model, the Thevenin equivalent voltage V_{th} at the relay is equal to the source voltage equal to 1.05 p.u. By running a load-flow analysis on the circuit model, the power factor is determined to be 0.96. The phase angle of $I_{prefault,phaseC}$ is -34° . Hence, using (4), angle θ_{vth} is calculated to be -18° . Using (5)–(8), the fault voltage at the relay is calculated to be

$$V_{monitor,phaseC} = 10.03 \angle -23.24^\circ \text{ kV.}$$

The actual fault voltage recorded by the relay is known to be

$$V_{actual,phaseC} = 9.98 \angle -28^\circ \text{ kV.}$$

The estimated fault voltage is very close to the actual voltage at the monitoring location. Next, the Takagi method is used to estimate the distance to the fault to be 4.99 mi. As mentioned earlier, the actual fault location is known to be 5.33 mi. Hence, using only the current waveform, the distance to fault has been estimated with 4.5% error (0.34 mi).

2) *Fault Location Using Current Magnitude:* For the analysis using only the current magnitude, pretend that the relay records only the fault current magnitude, as shown in Fig. 7.

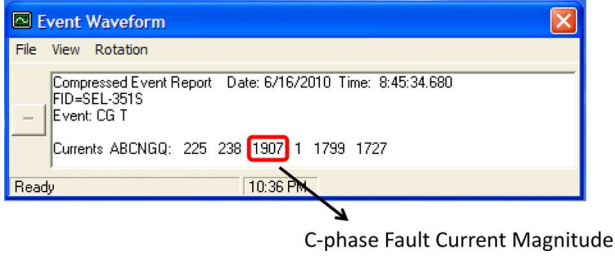


Fig. 7. Relay at the substation recording only the fault current magnitude.

The C-phase fault current magnitude recorded by the relay is 1907 A. As seen from Fig. 7, the currents flowing in the unfaulted phases are small compared to the fault current in phase C. Hence, ignoring the currents in the unfaulted phases, the sequence component of fault currents is determined by (14) as

$$|I_{f0}| = |I_{f1}| = |I_{f2}| = \frac{|I_c|}{3} = \frac{1907}{3} = 635.7 \text{ A.}$$

The actual sequence components of fault current recorded by the monitor are

$$\begin{aligned} |I_{f0, \text{actual}}| &= 583.1 \text{ A} \\ |I_{f1, \text{actual}}| &= 745.8 \text{ A} \\ |I_{f2, \text{actual}}| &= 554.4 \text{ A.} \end{aligned}$$

As seen before, the simplistic assumption in (11) results in an error of 9% when estimating I_{f0} and an error of 14.8% and 14.7% when estimating I_{f1} and I_{f2} , respectively.

Next, the equality constraint given by (20) is checked. Angle θ_{vth} is -24.81° while $(\theta_{z1} + \theta_{f1})$ is 2.65° . The difference of 27.47° is within the acceptable limit calculated in Section IV. Hence, (21)–(24) can be used to estimate the fault voltage at the relay. Thevenin voltage at the relay V_{th} is equal to the source voltage at 1.05 p.u. Using the magnitude of Z_1^{eq} and Z_0^{eq} , the magnitude of the missing phase-C fault voltage is calculated to be

$$V_{\text{monitor, phaseC}} = 9.81 \text{ kV.}$$

The actual magnitude of the fault voltage recorded by the relay is known to be

$$V_{\text{actual, phaseC}} = 9.98 \text{ kV.}$$

The error in estimation is around 1.23% with respect to the base voltage. To obtain the values of r_s and x_s , loop impedance in Ω/mi is calculated by (30) as

$$\begin{aligned} Z_n &= \frac{2z_{\text{line},1} + z_{\text{line},0}}{3} = 0.2577 + j1.0580 \Omega/\text{mi} \\ r_s &= 0.2577 \text{ and } x_s = 1.0580 \Omega/\text{mi.} \end{aligned}$$

The modified loop reactance method derived in (33) then estimates the distance to fault as 4.72 mi. The actual location is 5.33 mi. The error in estimation is 8% (0.61 mi).

B. Utility Test Case 2

The circuit model in this case study has been made available by a utility as part of the EPRI distribution fault-location study. The rated voltage at the substation is 34.5

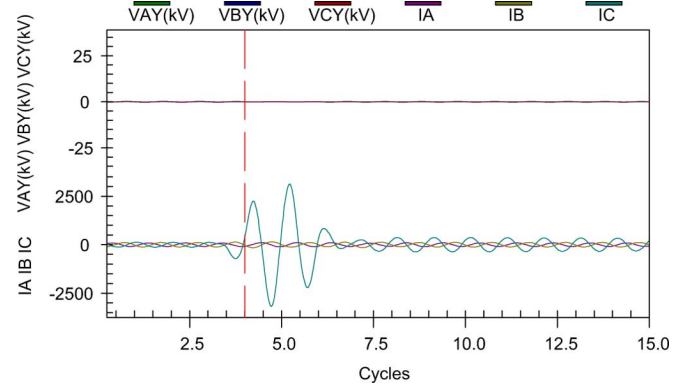


Fig. 8. Fault current and voltage recorded by the SEL-651 relay. Voltage data not usable.

kV. A SEL-651 relay located at the substation monitors the system current and voltage at a sampling rate of 32 samples/cycle. The Thevenin equivalent positive- and zero-sequence impedance upstream from the relay is nothing but the source impedance and is given as $Z_1^{eq} = 0.148 + j3.082 \Omega$ and $Z_0^{eq} = 0.00275 + j0.06103 \Omega$. The length of the main three-phase feeder is 5.79 mi and is constructed by using five different line configurations. The sequence impedance of the conductor used for constructing the majority length of the feeder (i.e., $z_{\text{line},1} = 0.1308 + j0.5546 \Omega/\text{mi}$ and $z_{\text{line},0} = 0.4029 + j1.8619 \Omega/\text{mi}$) has been utilized for computing the distance to fault. A phase-C fault has occurred in a recloser located 3.07 mi from the relay. No cause has been given. Fig. 8 illustrates the prefault and fault voltage and current waveform captured by the relay. This is the perfect example of a case where the relay fails to record the voltage at the monitoring location.

1) *Fault Location Using Current Phasors:* From the current waveform recorded by the relay, FFT operation is used to extract the prefault and fault current magnitude and phase angle of phase C. Since the fault persists for only two cycles, FFT operation has been performed on the second cycle. The currents are

$$\begin{aligned} I_{\text{prefault, phaseC}} &= 129 \angle -20^\circ \text{ A} \\ I_{\text{fault, phaseC}} &= 3200 \angle -79^\circ \text{ A.} \end{aligned}$$

To estimate the fault voltage phasor at the relay, the Thevenin voltage V_{th} at the monitoring location is assumed to be equal to the system source voltage (i.e., 1.04 p.u.). Since the circuit model of the system is available, a load-flow analysis shows that the power factor is 0.998. The phase angle of $I_{\text{prefault, phaseC}}$ is -20° . Hence, θ_{vth} is -16° , where θ_{vth} is the phase angle of Thevenin voltage V_{th} . Using (5)–(8), the fault voltage at the monitoring location is calculated to be

$$V_{\text{monitor, phaseC}} = 12.85 \angle -33^\circ \text{ kV.}$$

Using this estimated voltage, the Takagi method is then used to estimate the distance to the fault to be 3.06 mi. The actual fault distance is known to be 3.07 mi. Fault-location error is 0.2% (0.01 mi).

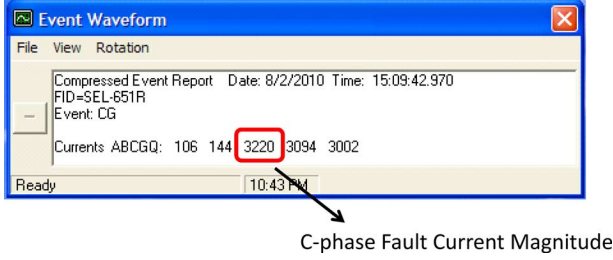


Fig. 9. Relay at the substation recording only fault current magnitude.

2) *Fault Location Using Current Phasors*: From the current magnitude recorded by the relay, as shown in Fig. 9, C-phase fault current magnitude is 3220 A. Ignoring the currents in the unfaulted phases, the sequence component of fault currents is determined by (14) as

$$|I_{f0}| = |I_{f1}| = |I_{f2}| = \frac{|I_c|}{3} = \frac{3220}{3} = 1073.3 \text{ A.}$$

The actual sequence fault currents are

$$\begin{aligned} |I_{f0,\text{actual}}| &= 1012.3 \text{ A} \\ |I_{f1,\text{actual}}| &= 1109.8 \text{ A} \\ |I_{f2,\text{actual}}| &= 1025.6 \text{ A} \end{aligned}$$

Since fault current is 23 times the current in the unfaulted phases, the estimated sequence currents are close to the actual sequence currents. In fact, the simplistic assumption in (11) results in an error of 6% when estimating I_{f0} and an error of 3.2% and 4.7% when estimating I_{f1} and I_{f2} , respectively. Next, the Thevenin voltage at the relay location V_{th} is assumed to be equal to the source voltage at 1.04 p.u. The equality constraint (20) cannot be verified since the voltage data is not available. The fault voltage at the relay is calculated using (21)–(24) to be

$$V_{\text{monitor,phaseC}} = 11.27 \text{ kV.}$$

To obtain the values of r_s and x_s , loop impedance in Ω/mi is calculated by (30) as

$$\begin{aligned} Z_n &= \frac{2z_{\text{line},1} + z_{\text{line},0}}{3} = 0.2215 + j0.9904 \Omega/\text{mi} \\ r_s &= 0.2215 \text{ and } x_s = 0.9904 \Omega/\text{mi.} \end{aligned}$$

The modified loop reactance method from (33) then estimates the distance to fault as 3.45 mi. The actual location is 3.07 mi. Hence, the error in estimation is 0.38 mi or 6.6%.

C. Summary of the Remaining Event Cases

Fault-location analysis on all seven cases has been summarized in Tables VI and VII. Utility test case 3 is a single momentary interruption consisting of three subevents, which represent the fast and slow operation of the recloser. For events 4–7, the actual location of the fault is unknown. Hence, for these events, to determine how accurate the location estimates are, faults were placed in the circuit model of the distribution feeder at the estimated distances to fault. Then, the fault current measured by the relay in the circuit model was compared to the fault current

TABLE VI
COMPARISON OF ACTUAL AND ESTIMATED DISTANCE TO FAULT USING CURRENT PHASORS AND CURRENT MAGNITUDE ONLY

Test Case No.	Actual Distance (miles)	Fault Location: Current Phasor		Fault Location: Current Magnitude	
		Distance Estimate (miles)	Error (miles)	Distance Estimate (miles)	Error (miles)
1	5.33	4.99	0.34	4.72	0.61
2	3.07	3.06	0.01	3.45	0.38
3a	2.67	2.71	0.04	2.93	0.26
3b	2.67	2.98	0.31	2.92	0.25
3c	2.67	2.83	0.16	2.96	0.29

TABLE VII
COMPARISON OF ACTUAL AND CIRCUIT MODEL FAULT CURRENT AT THE ESTIMATED DISTANCE TO FAULT

Test Case No.	Actual Distance (miles)	Actual Fault Current (kA)	Fault Location: Current Phasor		Fault Location: Current Magnitude	
			Distance Estimate (miles)	Circuit Model Fault Current (kA)	Distance Estimate (miles)	Circuit Model Fault Current (kA)
4	Unknown	13.14	0.02	13.28	0.03	13.14
5	Unknown	2.21	5.88	2.11	5.86	2.57
6	Unknown	4.14	2.29	4.47	2.60	4.11
7	Unknown	4.08	2.70	4.46	2.87	4.27

recorded by the relay in the real world. As seen from Table VII, the currents match closely and, hence, the location estimates are close to the actual fault location.

For the fault location using only the current magnitude, the assumption (11) made when calculating the sequence currents is evaluated for each event. As shown in Table VIII, fault current is much greater than twice the current in the unfaulted phases. Hence, the estimated sequence currents are close to the actual sequence currents, and the median of the errors is less than 5%. Next, the equality assumption given by (20) is checked in Table IX. For each event, θ_{vth} is not exactly equal to $(\theta_{z1} + \theta_{f1})$. However, the difference is within the 40° acceptable limit calculated in Section IV.

VI. CONCLUSION

This paper proposes practical impedance-based fault-locating methods using only current data recorded by a relay. Two fault-locating approaches using current phasors and current magnitude were developed. Both approaches work by estimating fault voltage at the monitoring location and then invoking impedance-based methods. Thevenin equivalent impedance upstream from the relay plays an important role in this voltage estimation and it is essential that this parameter be accurately calculated from the circuit model. Therefore, it is important to note that the circuit model must represent the actual feeder topology correctly. The erroneous circuit model will lead to inaccurate location estimates, simply because of errors in the circuit model.

TABLE VIII
ERROR BETWEEN THE ACTUAL AND ESTIMATED SEQUENCE FAULT CURRENTS

Fault Bus	Error %			Fault to Load Current Ratio
	I_{f0}	I_{f1}	I_{f2}	
1	9.0	14.8	14.7	8
2	6.0	3.3	4.7	23
3a	3.3	4.0	5.0	61
3b	0.6	1.3	1.8	64
3c	1.5	2.5	3.4	59
4	0.8	0.6	1.6	25
5	0.6	1.7	3.4	12
6	1.3	11.0	10.0	8
7	0.5	4.6	4.4	19

TABLE IX
EQUALITY CONSTRAINT CHECK FOR EACH FIELD DATA EVENT

Test Case No.	$\theta_{v_{th}}$ (degrees)	$\theta_{z1} + \theta_{f1}$ (degrees)
1	-24.81	2.65
2	Cannot be checked	
3a	-28.20	-18.83
3b	157.00	165.58
3c	-27.36	-14.29
4	-77.45	-66.42
5	185.00	210.44
6	107.59	133.53
7	32.79	47.92

Simulations of the algorithms on the modified IEEE 34-node test feeder and utility data showed promise. Fault location using only current phasors gave accurate estimates within 0.34 mi while estimates using fault location using only current magnitude were within 0.61 mi from the actual location of the fault. The analysis also concludes that performances of the algorithms are superior when fault current is greater than twice the load current.

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REFERENCES

- [1] *IEEE Guide for Determining Fault Location on AC Transmission and Distribution Lines*, IEEE Standard C37.114-2004, 2005.
- [2] K. Zimmerman and D. Costello, "Impedance-based fault location experience," in *Proc. 58th Annu. Conf. Protect. Relay Eng.*, Apr. 2005, pp. 211–226.
- [3] Elect. Power Res. Inst., "Distribution fault location: Field data and analysis," Palo Alto, CA, Tech. Rep., 2006.
- [4] G. Lampley, "Fault detection and location on electrical distribution system," in *Proc. IEEE Rural Elect. Power Conf.*, 2002, pp. B1–B15.
- [5] S. Das, S. Kulkarni, N. Karnik, and S. Santoso, "Distribution fault location using short-circuit fault current profile approach," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2011, pp. 1–7.
- [6] N. Kang and Y. Liao, "Fault location estimation using current magnitude measurements," in *Proc. IEEE SoutheastCon*, Mar. 2010, pp. 214–217.
- [7] Y. Liao, "Fault location using sparse current measurements," in *Proc. 39th North Amer. Power Symp.*, 2, 2007, pp. 1–6.
- [8] W. Kersting, "Radial distribution test feeders," in *Proc. IEEE Power Eng. Soc. Winter Meeting*, 2001, vol. 2, pp. 908–912.
- [9] S. Santoso, *Fundamentals of Electric Power Quality*, Winter ed. Scotts Valley, CA: CreateSpace, 2010.
- [10] W. D. Stevenson, Jr., *Elements of Power System Analysis*, 4th ed. New York: McGraw-Hill, 1982.
- [11] S. Kulkarni, N. Karnik, S. Das, and S. Santoso, "Fault location using impedance-based algorithms on non-homogeneous feeders," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2011, pp. 1–6.
- [12] M. A. Akcoglu, P. F. Bartha, and D. M. Ha, *Analysis in Vector Spaces Set*, 1st ed. Hoboken, NJ: Wiley, 2008.
- [13] N. Karnik, S. Das, S. Kulkarni, and S. Santoso, "Effect of load current on fault location estimates of impedance-based methods," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, Jul. 2011, pp. 1–6.

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